A

Mini Project Report

on

HUMAN WILDLIFE CONFLICT MONITORING & ALERT SYSTEM (WILD TRACK)

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on **WILD TRACK** has been submitted by **SANIYAH CHOGLAY (24107021), VISHWAS PRAJAPATI (24107057)**, **SHWETALI PHADATRE(24107136)** who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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ACKNOWLEDGEMENT

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Chapter 1

1. Introduction

Human–wildlife conflict is a serious issue in many regions where human settlements are located close to forests and wildlife habitats. Due to increasing population, urban expansion, and deforestation, wildlife animals often enter human areas in search of food and shelter. This leads to conflicts between humans and animals, resulting in damage to crops, loss of livestock, property destruction, and sometimes injuries or death to humans.

India is one of the countries that frequently experiences human–wildlife conflicts. Animals such as elephants, leopards, wild boars, and monkeys often move into villages and agricultural fields. These incidents create fear among local communities and require immediate attention from forest authorities.

Traditional systems for reporting wildlife sightings rely on manual communication with forest officials or local authorities. These methods are often slow and inefficient, which can delay response time and increase the risk of dangerous encounters.

With the advancement of technology, mobile applications can play an important role in monitoring wildlife activity and alerting nearby communities. Mobile devices equipped with GPS and internet connectivity can be used to report wildlife sightings in real time and notify other users in nearby areas.

Inspired by the WildWatch mobile application developed in Kerala, this project proposes the development of a Human Wildlife Conflict Monitoring and Alert System Mobile Application. The application allows users to report wildlife sightings, upload images, and share location information. Based on these reports, the system sends alerts to nearby users and authorities. The proposed system aims to improve public safety, reduce wildlife conflict incidents, and help forest authorities monitor wildlife movement more efficiently.

1.1 Purpose

The purpose of this project is to design and develop a mobile application that helps monitor and manage human–wildlife conflicts by providing a platform for reporting wildlife sightings and sending alerts to nearby users.

This system aims to improve communication between citizens and forest authorities by enabling real-time information sharing about wildlife presence in specific locations. By using location-based technology, the application can notify people in nearby areas about potential wildlife threats.

The application also allows users to upload photographs and descriptions of wildlife sightings. This information can help forest officials verify incidents and take necessary actions quickly.

The main purpose of the project report is to describe the development process, system design, and implementation details of the Human Wildlife Conflict Monitoring and Alert System.

1.2 Problem Statement

Human–wildlife conflict is increasing due to rapid urbanization, deforestation, and expansion of human activities into wildlife habitats. People living near forests often face unexpected encounters with wild animals, which can lead to dangerous situations.

In many cases, there is no quick and reliable method to inform nearby communities about wildlife movement. Traditional reporting methods such as phone calls or manual reports to forest departments take time and may not reach all people who are at risk.

Because of the lack of real-time information, villagers and travelers may unknowingly enter areas where wild animals are present. This increases the risk of accidents and panic situations.

Therefore, there is a need for a technological solution that can provide real-time wildlife monitoring and alert systems. A mobile application can help collect wildlife sighting data and notify nearby users instantly.

The proposed system addresses this problem by providing a platform for reporting wildlife sightings and sending alerts to users based on their geographic location.

1.3 Objectives

The main objectives of the Human Wildlife Conflict Monitoring and Alert System are:

- To develop a mobile application that allows users to report wildlife sightings easily.
- To provide real-time alerts and notifications to nearby users when wildlife activity is reported.
- To enable users to upload images and location details of wildlife sightings.
- To maintain a centralized database for storing wildlife reports and related information.
- To assist forest authorities in monitoring wildlife movement patterns.
- To improve communication between local communities and forest departments.
- To increase public awareness about wildlife presence and safety precautions.
- To reduce the number of human–wildlife conflict incidents.

1.4 Scope

The scope of the Human Wildlife Conflict Monitoring and Alert System includes the development of a mobile-based platform for reporting wildlife sightings and providing alerts to users in nearby areas.

The system will allow users to create reports that include information such as the type of animal, location, time, and optional images. This data will be stored in a centralized database and used to generate alerts.

Forest officials can use the collected data to monitor wildlife activity and take preventive measures in high-risk areas.

Although the current system focuses on mobile-based reporting and alerts, future improvements may include advanced technologies such as artificial intelligence for wildlife detection, integration with sensor networks, and predictive analysis of wildlife movement.

Chapter 2

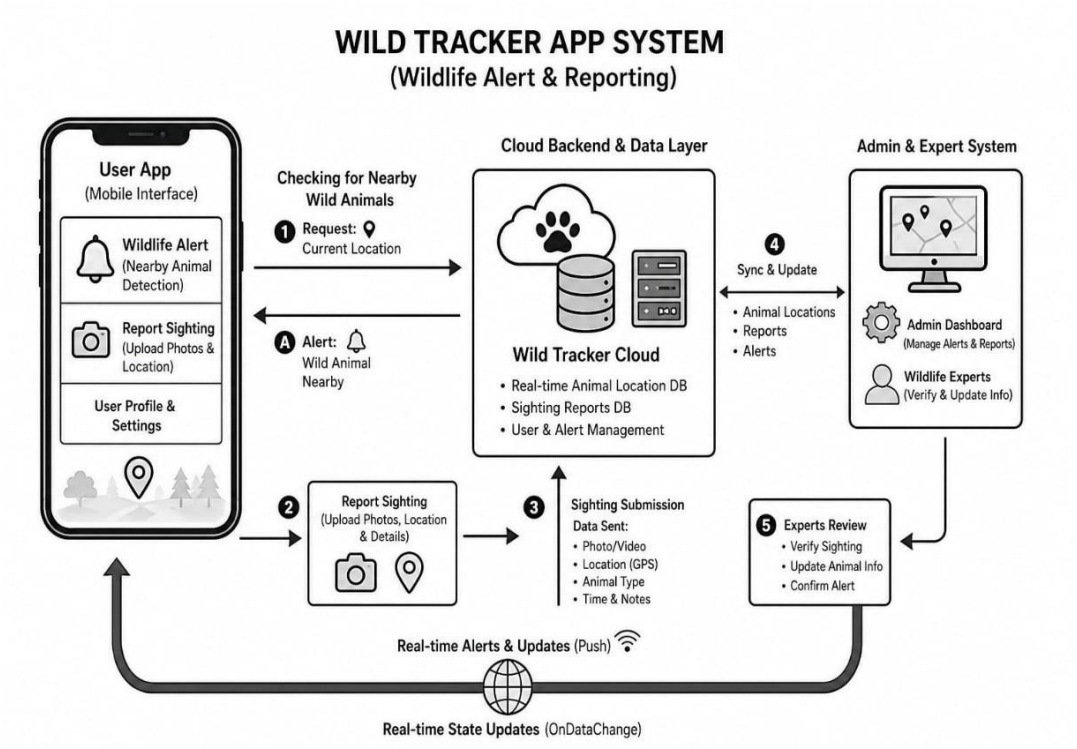
2. Proposed System

The proposed system is a mobile application designed to monitor wildlife activity and reduce human–wildlife conflicts by providing real-time reporting and alert services.

The application allows users to report wildlife sightings directly from their mobile devices. When a report is submitted, the system stores the information in a database and sends alerts to nearby users.

The system also allows forest authorities to view reports and analyze wildlife movement patterns. By using GPS-based location tracking, the application can identify areas where wildlife activity is frequently reported.

This system improves safety by providing early warnings and helps authorities respond quickly to wildlife incidents.



2.1 Features and Functionality

The main features of the proposed system include:

- **User Registration and Login:**
  Users must create an account to access the application features.
- **Wildlife Sighting Reporting:**
  Users can report wildlife sightings by entering details such as animal type, location, and description.
- **GPS Location Tracking:**
  The system automatically captures the user's location when a report is submitted.
- **Real-Time Alerts:**
  Nearby users receive notifications when wildlife sightings are reported.
- **Admin Monitoring:**
  Forest authorities can view and manage wildlife reports through an admin interface.
- **Data Storage:**
  All wildlife reports are stored in a database for future analysis.

Chapter 3

The successful implementation of this project is expected to bring significant improvements in managing human–wildlife conflicts. It focuses on providing a simple and effective way for people to stay informed about wildlife activity in their surroundings. By using modern mobile technology, the system makes it easier to report sightings and share important information quickly. This helps in building awareness and reducing panic in risky situations. The collected data can also support better decision-making by authorities. Overall, the project aims to create a safer environment for both humans and wildlife through timely communication and monitoring.

3. Project Outcomes

1. The expected outcomes of the project include the successful development of a mobile application that helps monitor wildlife activity and reduce human–wildlife conflicts.
2. The system will provide a platform where users can report wildlife sightings quickly and easily. This will help create a database of wildlife movement patterns in different regions.
3. The alert system will help people stay informed about wildlife presence in their surroundings, which can reduce the risk of dangerous encounters.
4. The project will also demonstrate how mobile technology can be used for environmental monitoring and public safety.

Chapter 4

The development of this project requires a set of software tools and technologies to ensure smooth functioning of the system. These requirements help in building, testing, and running the mobile application effectively. Choosing the right software components is important for achieving better performance and reliability. The system is designed to work on widely used platforms so that it is easily accessible to users. It also depends on proper development tools and database support for storing and managing data. Overall, these software requirements form the foundation for developing a stable and efficient application.

4. Software Requirements

The software requirements for developing the Human Wildlife Conflict Monitoring and Alert System include:

A. Operating System
B. Android or iOS mobile operating systems.
C. Programming Languages
D. Java, Kotlin, or Flutter for mobile app development.
E. Database
F. Firebase or MySQL for storing wildlife reports and user data.
G. Development Tools
H. Android Studio or Visual Studio Code.
I. Internet Connectivity
J. Required for sending and receiving wildlife reports and alerts.

Chapter 5

5. Project Design

Project design defines the architecture, workflow, and structural components of the Human Wildlife Conflict Monitoring and Alert System. The system is designed as a mobile-based platform that allows users to report wildlife sightings and receive alerts about wildlife activity in nearby areas.

The design of the system focuses on three main components: the mobile application interface, the backend server with database storage, and the administrative monitoring system. These components work together to ensure smooth data flow and effective communication between users and forest authorities.

The system architecture ensures that wildlife sighting reports are collected from users, stored securely in a database, and then processed to generate alerts for nearby users. The design also includes location tracking using GPS to identify the exact area of wildlife activity.

The project design includes the following diagrams:

- System Architecture Diagram
- Use Case Diagram
- Data Flow Diagram (DFD)
- Entity Relationship (ER) Diagram

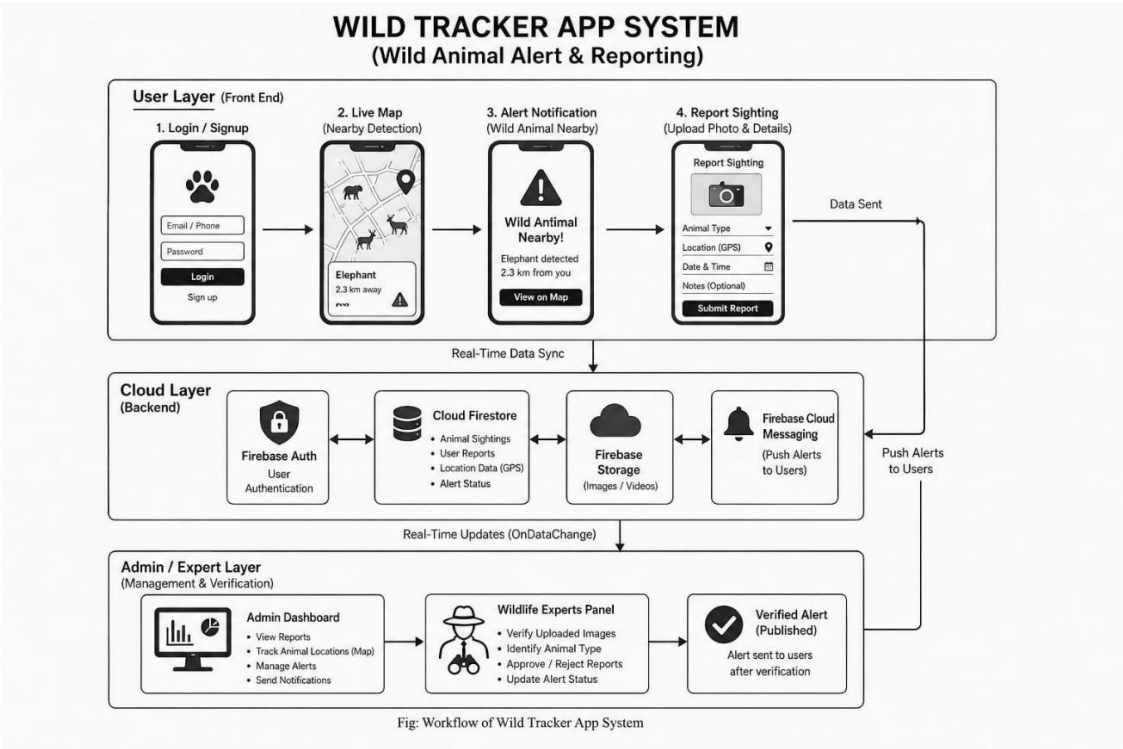


Figure 5.1: Workflow Diagram

5.1 System Architecture

The system architecture represents the overall structure of the Human Wildlife Conflict Monitoring and Alert System. It shows how different components interact with each other.

The architecture consists of three main layers:

User Layer
The user layer includes the mobile application used by citizens. Through this application, users can register, log in, report wildlife sightings, upload images, and receive alert notifications.

Application Layer
The application layer handles the processing of user requests. When a wildlife sighting is reported, the application collects information such as location, image, and description. This data is then sent to the server.

Database Layer
The database layer stores all information related to users, wildlife reports, images, and location data. The stored data can be accessed by administrators and used to analyze wildlife movement patterns.

When a user reports a wildlife sighting, the system processes the information and sends alerts to other users located near the reported area. This architecture ensures quick communication and effective wildlife monitoring.

5.2 Use Case Diagram

A Use Case Diagram describes the interaction between users and the system. It identifies the different actions that users can perform in the application.

The system mainly has two actors:

User:
The user is a citizen who uses the mobile application to report wildlife sightings and receive alerts.

Admin:
The admin represents forest authorities who monitor reports and manage the system.
Main use cases include:

- User registration
- User login
- Reporting wildlife sighting
- Uploading wildlife images
- Viewing alerts
- Admin monitoring reports
- Admin managing users

These interactions help the system function effectively by allowing users to contribute information and administrators to supervise wildlife monitoring activities.

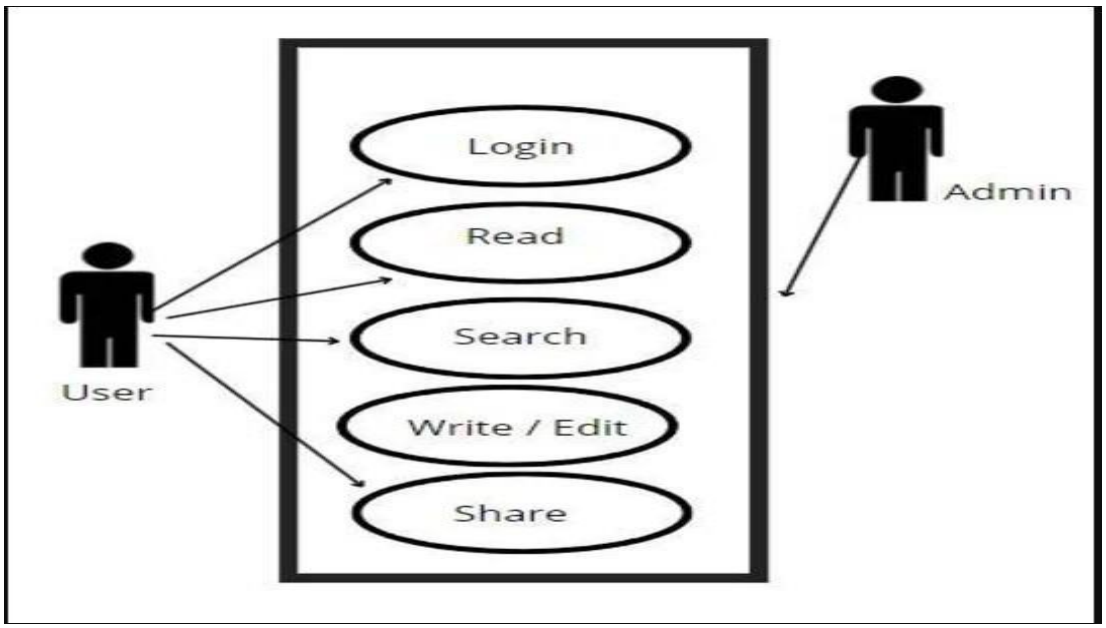


Figure 5.2: Use Case Diagram of Wildlife Monitoring System

5.3 Data Flow Diagram (DFD)

A Data Flow Diagram shows how information moves through the system.

The main processes in the system include:

User Input:
The user enters wildlife sighting details such as animal type, location, and description.

Data Processing:
The system processes the information and verifies the location using GPS.

Data Storage:
The information is stored in the central database.

Alert Generation:
The system sends notifications to nearby users and authorities.

This diagram helps understand how data flows from the user to the system and how alerts are generated.

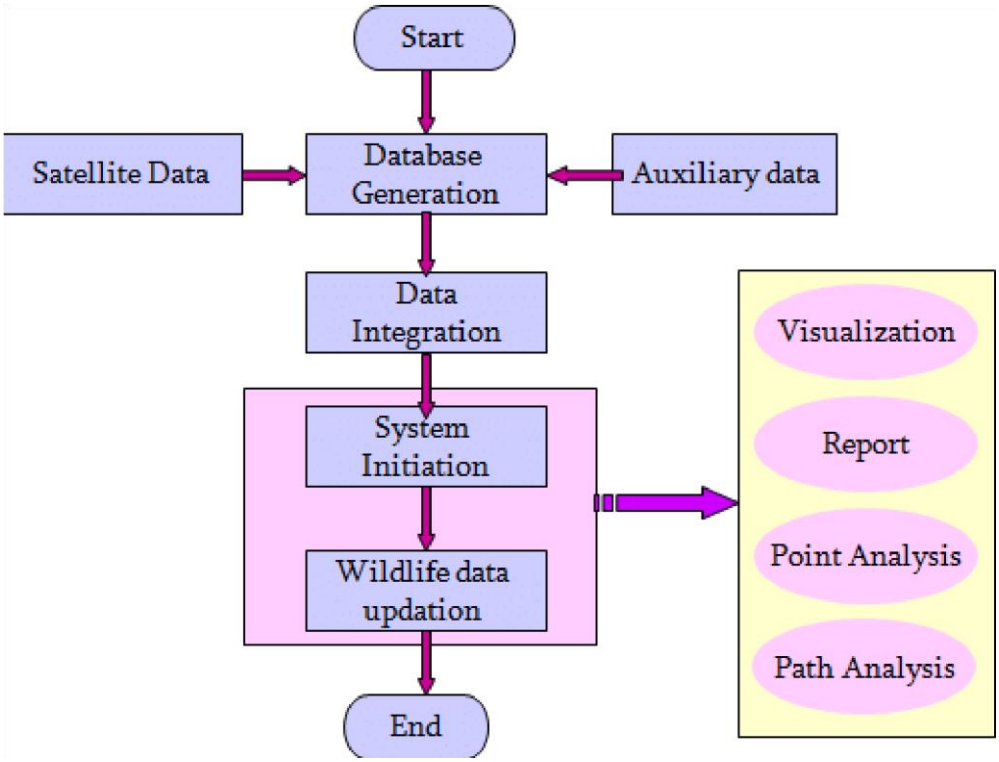


Figure 5.3: Data Flow Diagram of Proposed System

5.4 Entity Relationship Diagram (ER Diagram)

The ER Diagram represents the database structure of the system. It shows the relationship between different data entities stored in the database.

Important entities include:

User
Stores user information such as user ID, name, email, and password.

Wildlife Report
Stores details of wildlife sightings including animal type, location, time, and description.

Image
Stores photos uploaded by users during wildlife reporting.

Alert
Stores alert notifications sent to users.

The relationships between these entities help maintain organized and structured data storage.

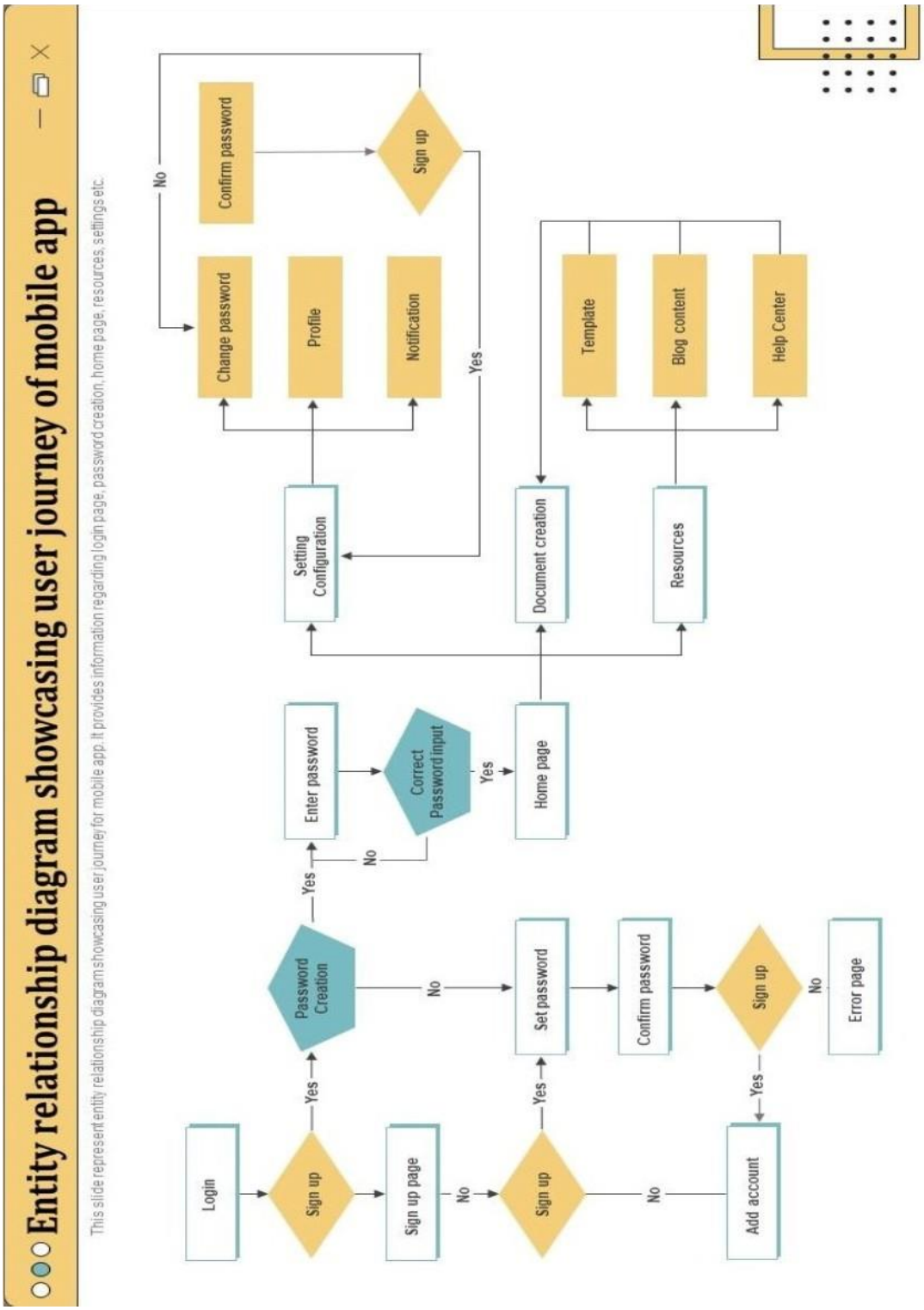


Figure 5.4: Entity Relationship Diagram

Chapter 6

6. Project Scheduling

development. Proper scheduling is important to ensure that each phase of the project is completed on time and in an organized manner. It helps in dividing the work into smaller tasks and assigning timelines This section describes the planning and scheduling of the project activities throughout its for each step. By following a structured schedule, the team can track progress and manage resources effectively. This approach ensures smooth execution of the project from start to completion.

PROJECT TITLE: HUMAN WILDLIFE CONFLICT MONITORING & ALERT SYSTEM
PROJECT GUIDE: DR. AVINASH
INSTUTUTE & DEPARTMENT NAME: AP SHAH INSTITUTE OF TECHNOLOGY(ICSE Data Science)
DATE: 4-1-26

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION (Weeks)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization, Identifying the scope and objectives of the Mini Project	VISHVAS PRAJAPATI	1-13-26	1-16-26	1	100%
1.2	Identifying the functionalities of the Mini Project	SANYAH CHOGLEY	1-16-26	1-20-26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	SHVETALI PHADATARE, SANYAH CHOGLAY, VISHVAS PRAJAPATI	1-20-26	2-3-26	2	100%
1.4	Designing the Graphical User Interface(GUI)	SHVETALI PHADATARE	2-3-26	2-17-26	2	100%
1.5	Presentation I	SHVETALI PHADATARE, SANYAH CHOGLAY, VISHVAS PRAJAPATI	2-17-26	2-24-26	1	100%
2	**Project Design and Implementation**					
2.1	Database Design	SHVETALI PHADATARE, SANYAH CHOGLAY, VISHVAS PRAJAPATI	2-24-26	3-10-26	2	100%
2.2	Database Connectivity of all modules	SANYAH CHOGLEY	3-10-26	3-17-26	1	100%
2.3	Integration of all modules and Report Writing	SHVETALI PHADATARE	3-17-26	3-24-26	1	100%
2.4	Presentation II	SHVETALI PHADATARE, SANYAH CHOGLAY, VISHVAS PRAJAPATI	3-24-26	4-1-26	1	100%

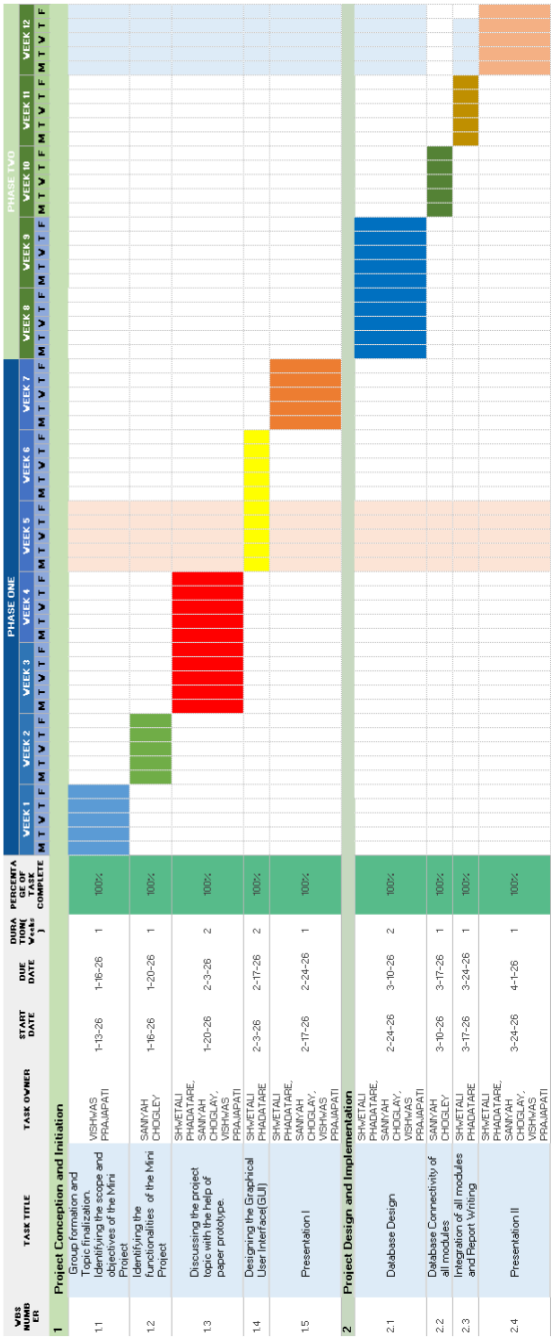


Table 6.1: Project Gantt Chart

Chapter 7

7. Results

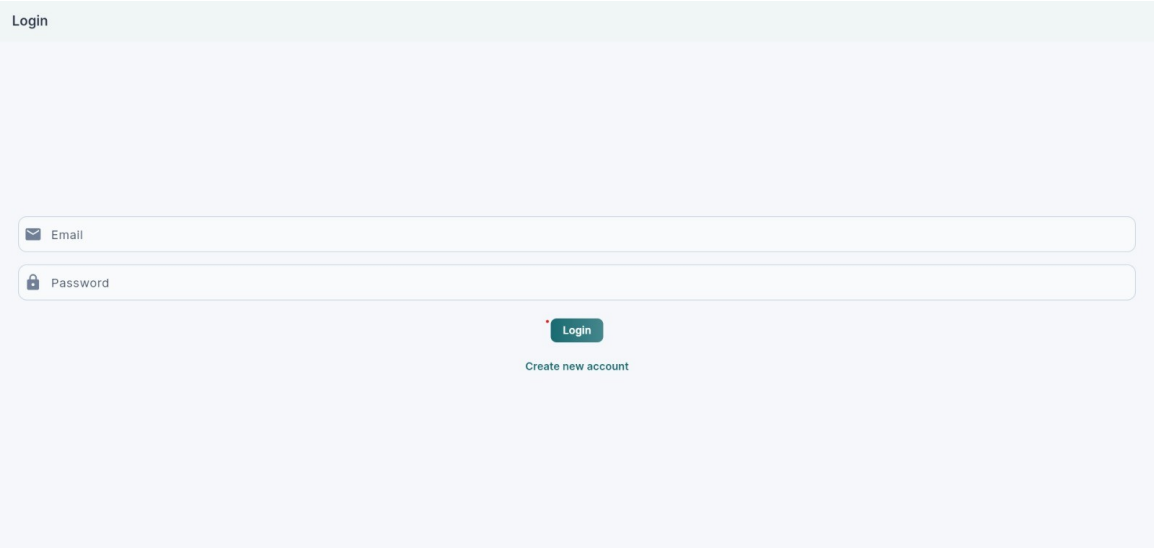
The Human Wildlife Conflict Monitoring and Alert System was successfully developed and tested. The mobile application demonstrated the ability to collect wildlife sighting reports from users and generate alerts for nearby communities.

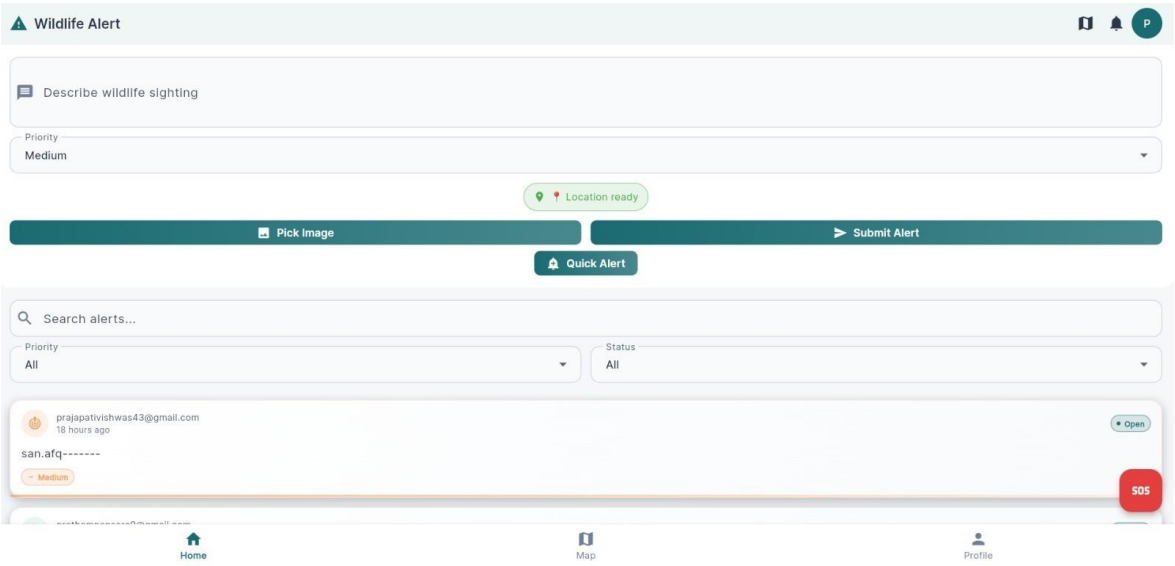
The application allowed users to create accounts, log in, and submit wildlife sighting reports along with images and GPS location data. These reports were stored in the database and displayed in the application interface.

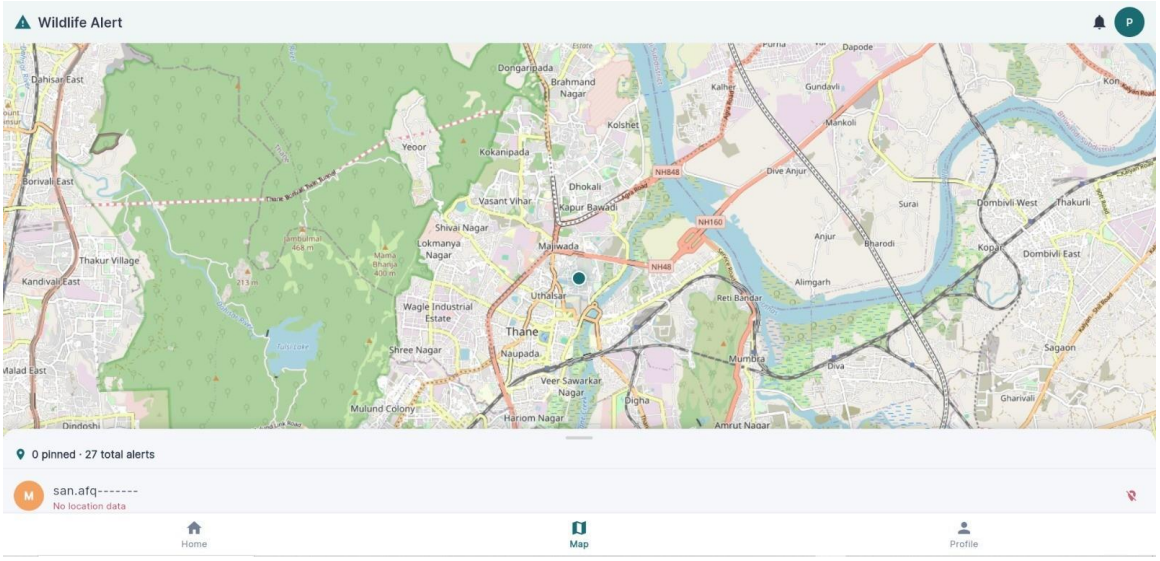
The alert notification system worked effectively by notifying users when wildlife activity was reported near their location. This feature can help people avoid dangerous areas and take necessary precautions.

Testing results showed that the system can operate efficiently with realtime data input and provide useful information to both users and administrators.

The project results demonstrate that mobile technology can be effectively used for environmental monitoring and wildlife conflict management.






Wildlife Alert
P
Dahisar East
Borivali East
Thakur Village
Kandivali East
Dongaripada
Brahmand Nagar
Kolshet
Yeoor
Kokanipada
Dhokali
Vasant Vihar
Kapur Bawadi
Shivai Nagar
Lokmanya Nagar
Majiwada
Uthalsar
Wagle Industrial Estate
Thane
Shree Nagar
Naupada
Veer Sawarkar Nagar
Mulund Colony
Hariom Nagar
Digha
Kalher
Gundavli
Mankoli
Dive Anjur
Surai
Anjur
Bharodi
Alimgarh
Reti Bandar
Mumbra
Diva
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Kopar
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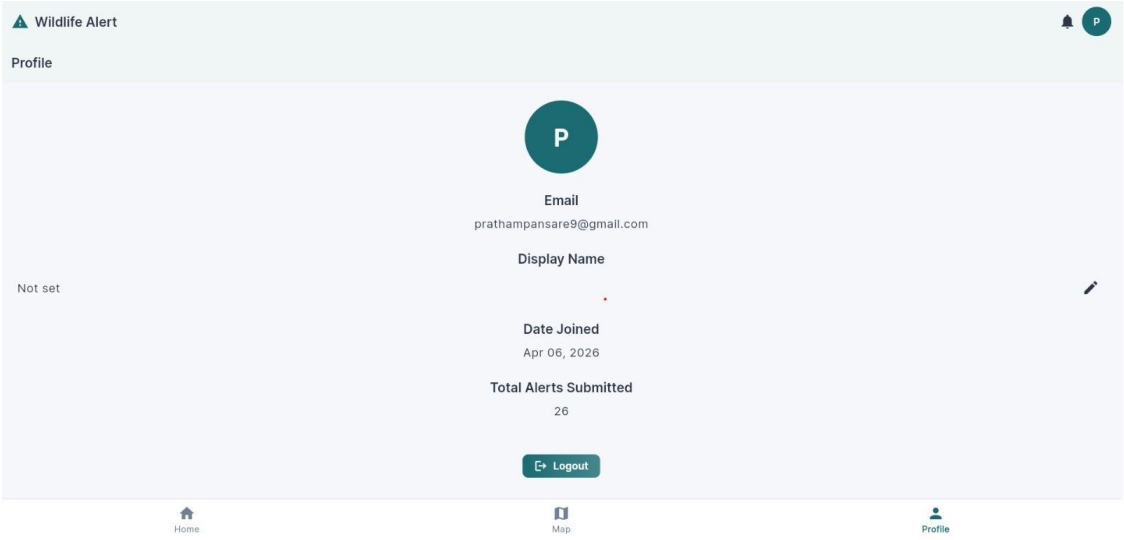
Wildlife Alert
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Profile
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Email
prathampansare9@gmail.com
Display Name
Not set
Date Joined
Apr 06, 2026
Total Alerts Submitted
26
Logout
Home
Map
Profile

CHAPTER 8

8. Conclusion

Human–wildlife conflict has become an increasing problem in regions where human settlements are located close to forests, creating serious risks for both people and animals. To address this issue, the Human–Wildlife Conflict Monitoring and Alert System offers a technological solution designed to reduce such conflicts through real-time reporting and alert mechanisms. The mobile application developed as part of this project enables users to report wildlife sightings, share their location, and receive notifications about wildlife activity in nearby areas. By enhancing communication between citizens and forest authorities, the system helps minimize the chances of dangerous encounters while also supporting wildlife conservation efforts. Overall, this project highlights how mobile technology and location-based services can be effectively used to build a reliable and efficient wildlife monitoring system.

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